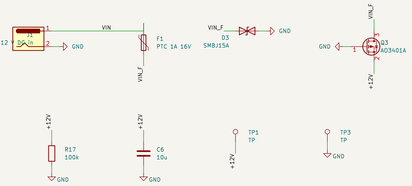
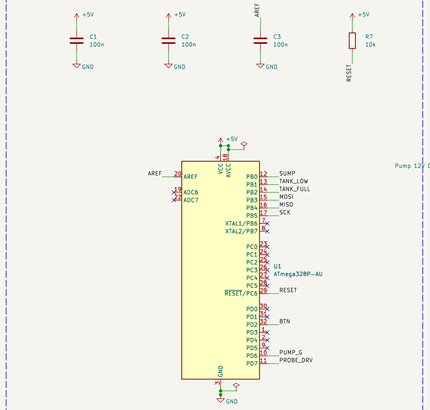
12 V INPUT: jack, PTC, TVS, reverse-polarity PMOS



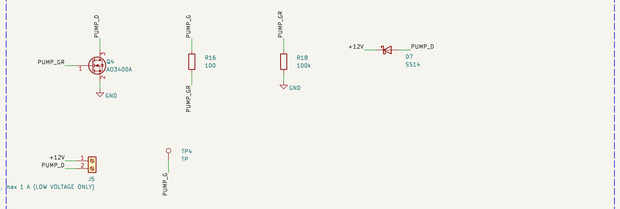
5 V REGULATOR



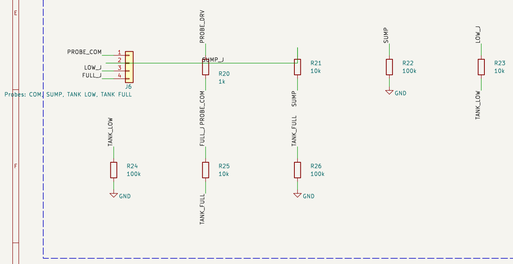
MCU: ATmega328P-AU, 8 MHz internal



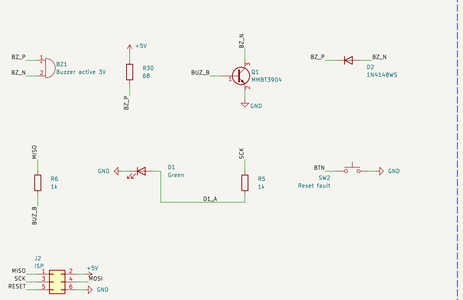
PUMP DRIVER (12 V DC, low voltage only): logic-level NMOS, flyback diode, terminal



AC-EXCITED WATER PROBES: common drive through 1 k, each sense 10 k series, 100 k pull-down



ALARM, STATUS, PROGRAMMING



Jal-Rakshak LOW-VOLTAGE prototype board (never mains). 12 V DC in through PTC, TVS and reverse-polarity PMOS; AMS1117-5.0 for logic. The probe common electrode is driven by D7 through 1 k; a probe reads wet only if its sense pin follows the drive HIGH and LOW (AC excitation, no electrolysis). Pump: 12 V DC mini pump up to 1 A on J5, low-side AO3400A with SS14 flyback. Status LED on D13 and active buzzer on D12 share the ISP lines as on an Arduino Nano. Firmware: products/04-jal-rakshak/firmware (8 MHz build).

First engineering release; see pcb/README.md for status and limits

ClaudeWork	
Sheets: /	
File: jal_rakshak.kicad_sch	
Title: Jal-Rakshak tank controller — low-voltage prototype board (NOT for m	
Size: A1	Date: 2026-11-01
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4	**Rev: A**
	Id: 1/1